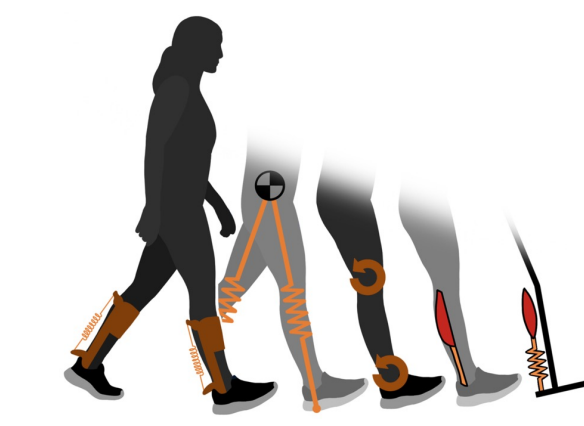




Why Does Wearing High Heeled Footwear Stiffen the Achilles Tendon?

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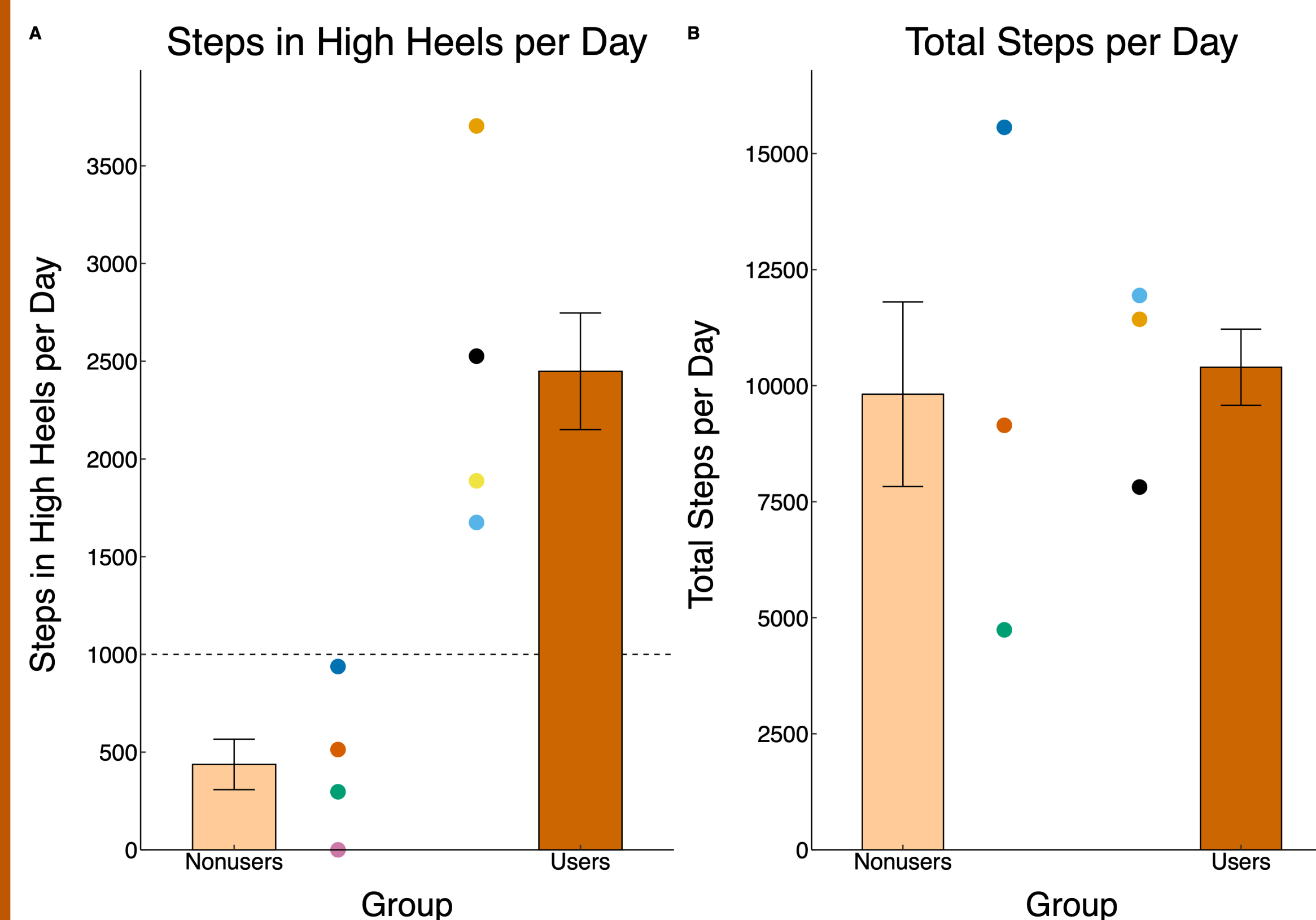
Elevated Heels = Stiffer Tendons

- Habitually wearing high heels increases Achilles tendon stiffness [1]
- Tendon strain may drive tendon stiffness remodeling [2]

Research Question: Does tendon strain explain the increased tendon stiffness due to wearing high heels?

Protocol

- 8 Participants**
 - Not habitual heel wearers prior to participation
 - Users wore high heels >1500 steps/day (n=4)
 - Nonusers wore for <1000 steps/day (n=4)



- Custom high heels:**
 - Set the ankle at 14° plantar flexion
 - Provided to participants for 14 weeks

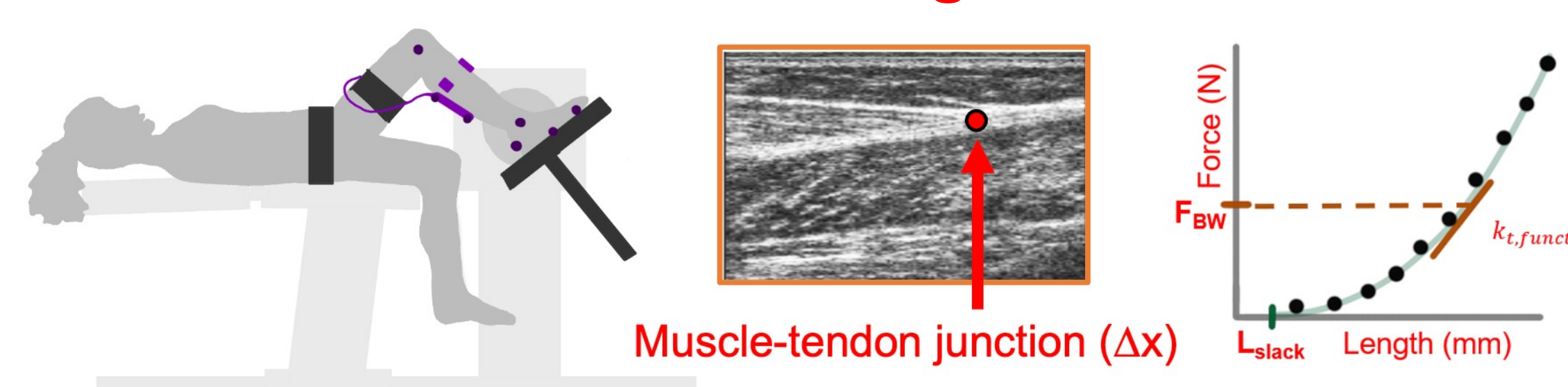


- Tendon stiffness was measured using ultrasonography and dynamometry [3]
- We determined walking biomechanics and estimated Achilles tendon strain during treadmill walking in flats and high heels

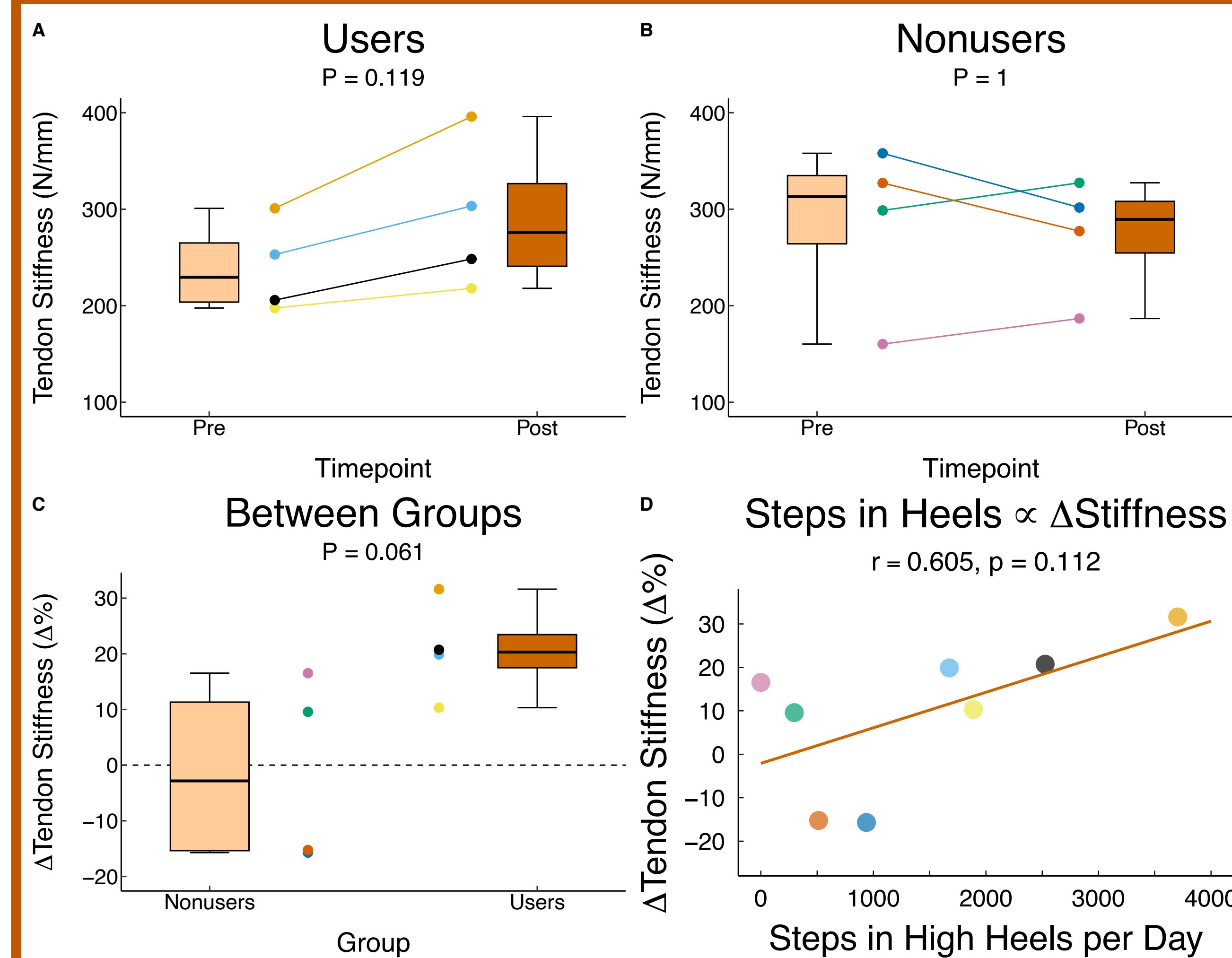
Dynamometer Testing

Ultrasound Image

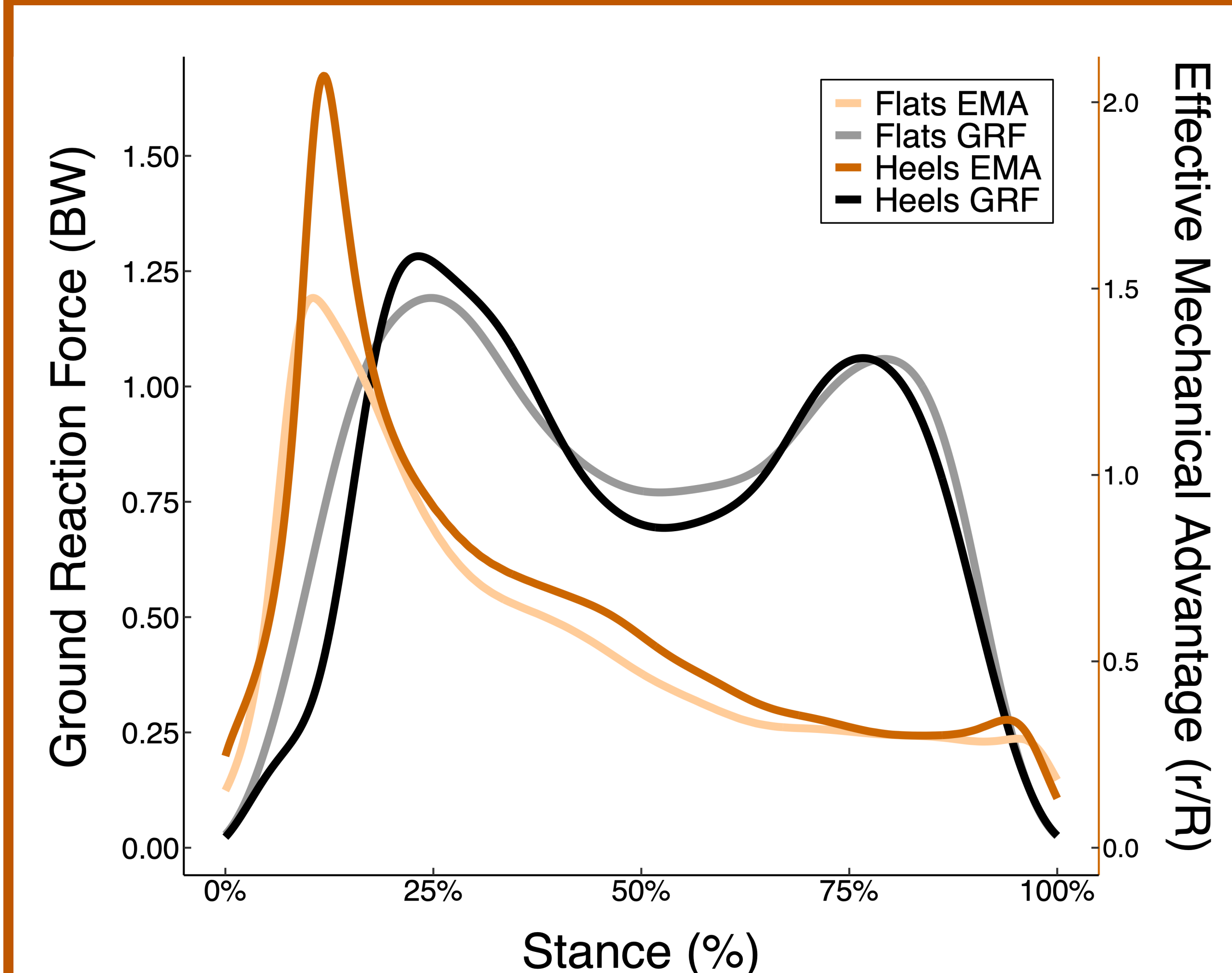
Tendon F-L Profile



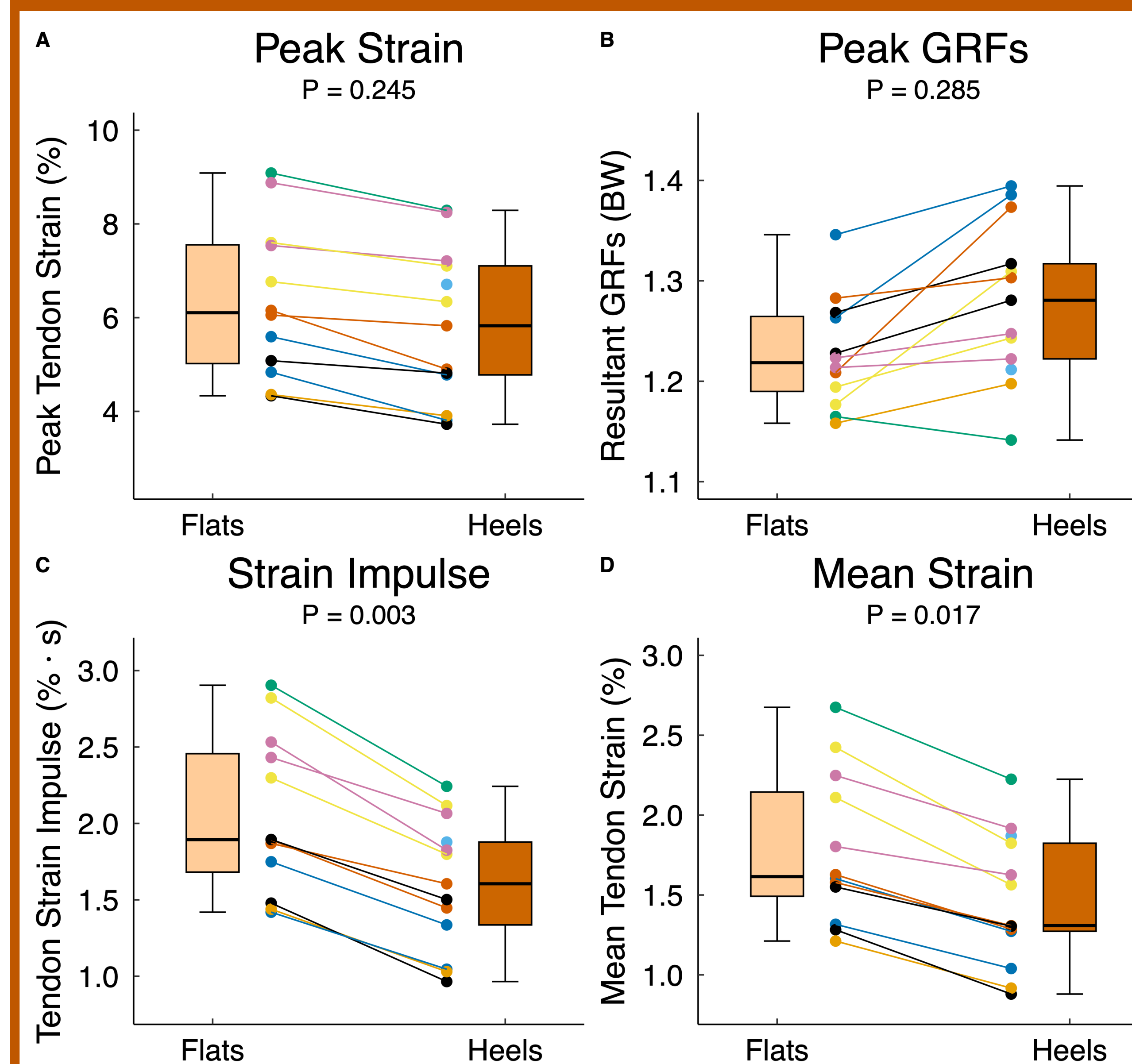
Wearing high heels increased Achilles tendon stiffness



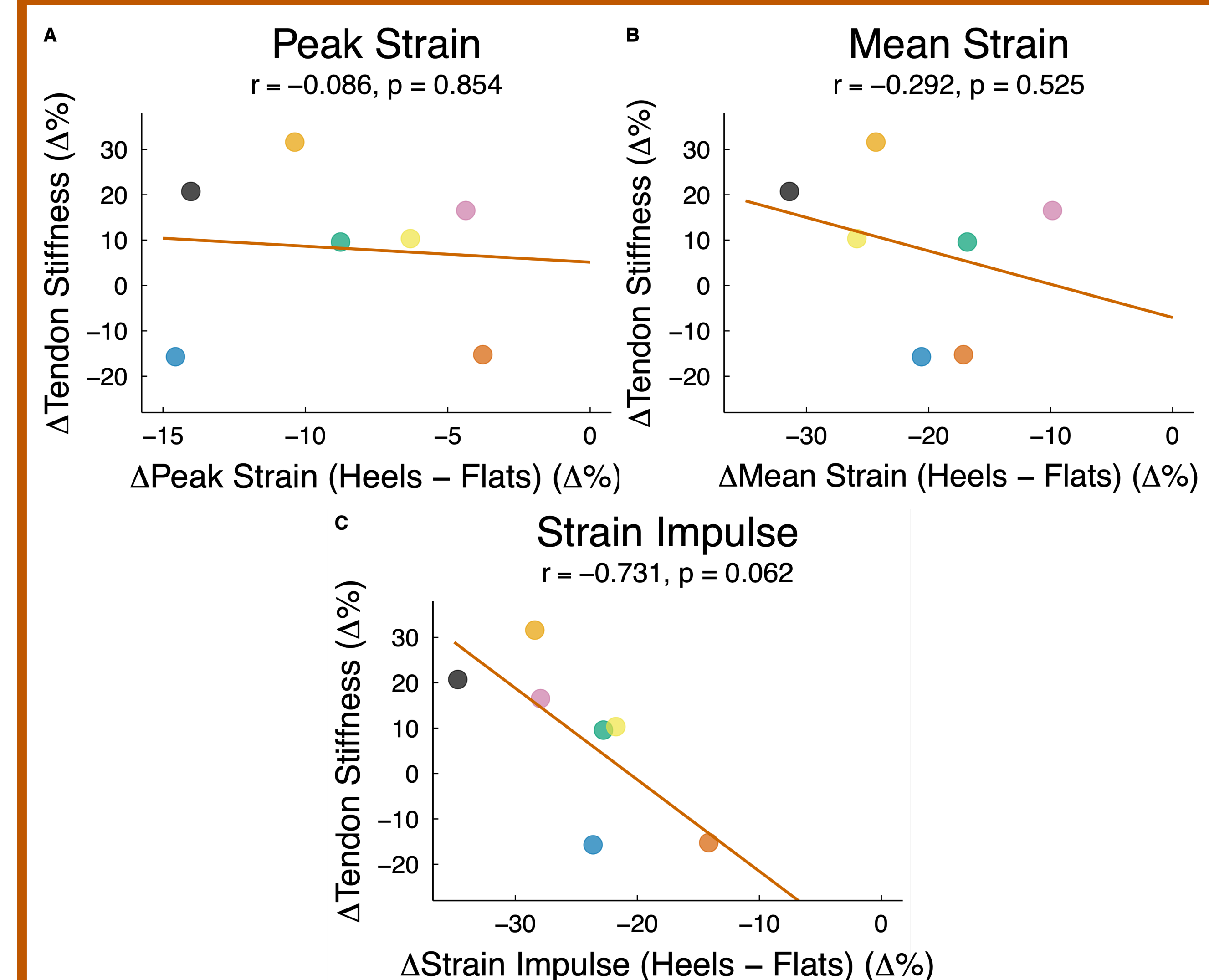
Increased ankle effective mechanical advantage explains lower tendon strain



Walking in high heels did NOT increase tendon strain



Strain magnitude did NOT drive tendon stiffness changes



The mechanism(s) underlying the Achilles tendon stiffening due to high heel use remain elusive

- High heels stiffen the Achilles tendon despite reducing Achilles tendon strain

Acknowledgements & References

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- Csapo, J Exp Biol 2010
- McMahon, J Str Cond Res 2022
- Beck et al. J Appl Physiol 2024